

Experiment No.	4
Experiment Title	Binary Classification using Linear and Kernel-Based Models
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1 Objective

The objective of this experiment is to classify emails as spam or ham using two different types of classifiers, Logistic Regression (a linear, probability-based model) and Support Vector Machine (a margin-based model that can use different kernels). Both models are tuned using grid search and compared on the same test set, so their strengths and limitations can be judged directly against each other rather than in isolation.

2 Problem Statement

Given a set of numeric features extracted from an email (how often certain words and characters appear, and the length of runs of capital letters), the task is to predict whether the email is spam (1) or ham (0). This is a binary classification problem. The input is a row of 57 numeric features per email, and the output is a single class label.

3 Dataset Description

Dataset Name	Spambase
Dataset Source	UCI Machine Learning Repository (also mirrored on Kaggle)
Number of Samples	4,601 emails originally; 4,210 after removing 391 duplicate rows during preprocessing
Number of Features	57 numeric features (48 word frequencies, 6 character frequencies, 3 capital-letter run-length features)
Number of Classes	2 (spam = 1, ham = 0)
Missing Values	None
Train-Test Split	80:20, stratified so both sets keep the same 60:40 ham-to-spam ratio

4 Exploratory Data Analysis

This section looks at the dataset overview, class balance, correlation between features, and which individual features are most linked to the spam label.

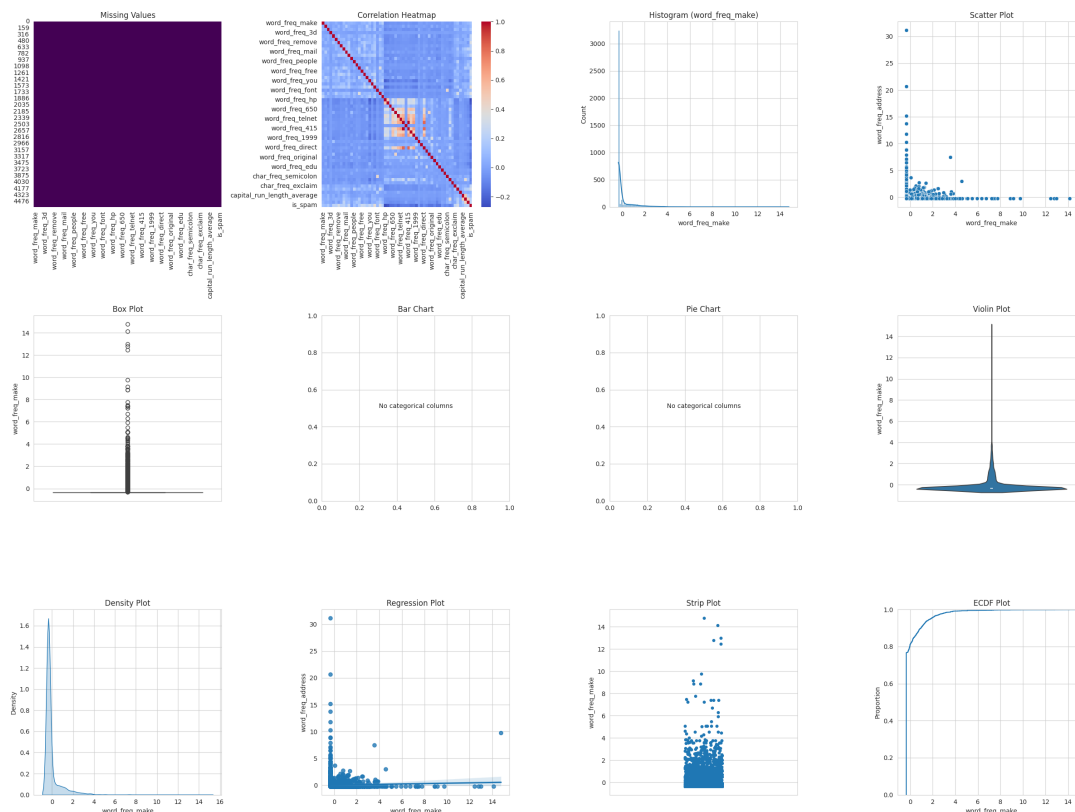


Figure 1: General EDA grid: missing values, correlation heatmap, and distribution plots

Inference:

The missing value panel confirms the dataset has no gaps to fill in. The correlation heatmap is dense since there are 57 features, but it is still useful for spotting groups of related word-frequency columns that tend to move together. The histogram, box and violin plots (built on the first feature, word_freq_make) show a heavily right-skewed distribution, most emails rarely or never contain that particular word, which is typical for this kind of frequency-based feature.

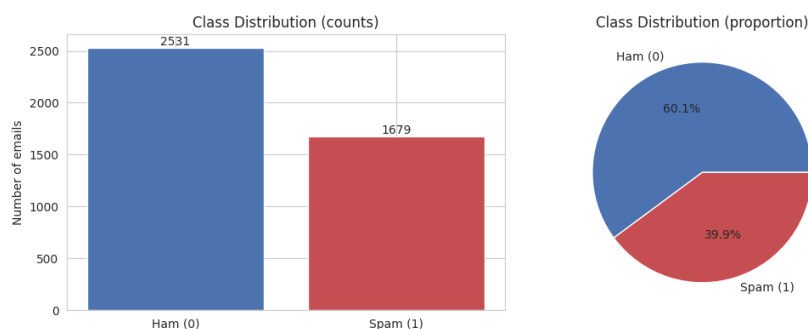


Figure 2: Class distribution: spam vs ham

Inference:

The dataset is moderately imbalanced, about 60% ham and 40% spam. This is not extreme, but it is skewed enough that accuracy alone would not tell the full story, which is why precision, recall and F1-score are also reported later rather than accuracy on its own.

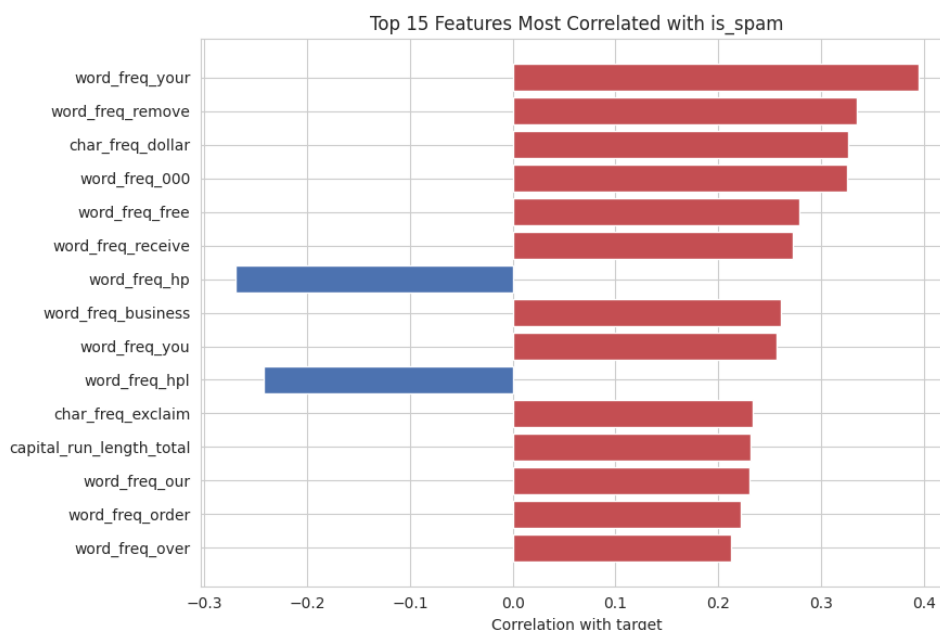


Figure 3: Top 15 features most correlated with the spam label

Inference:

Words and characters commonly seen in promotional or scam-style emails, such as `word_freq_your`, `word_freq_remove`, `word_freq_free` and `char_freq_dollar`, show the strongest positive correlation with the spam label. This matches intuition about what spam usually contains and confirms the dataset has a learnable signal before any model is trained.

5 Data Preprocessing

- **Missing value handling:** not required here since the dataset has no missing values, though the preprocessing function checks for and would handle them automatically.
- **Encoding:** not required either, every feature and the target are already numeric.
- **Feature scaling:** all 57 feature columns were standardized (mean 0, standard deviation 1) using `StandardScaler`, which matters for both models used here, Logistic Regression's regularization penalty and SVM's margin calculation are both sensitive to the scale of the input features.
- **Train-test split:** an 80:20 split was used, stratified on the target label so the class ratio stays consistent in both sets.
- **Feature engineering:** none was needed, the dataset already provides pre-extracted numeric features rather than raw email text.
- **Duplicate removal:** 391 duplicate rows were found and removed during preprocessing, bringing the working dataset down from 4,601 to 4,210 rows.

6 Machine Learning Algorithm

Logistic Regression. Working principle: estimates the probability that an email is spam using the sigmoid function applied to a weighted sum of the input features, then applies a 0.5 threshold to get a class label. Mathematical formulation:

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-(w^T x + b)}}$$

Advantages: fast to train, easy to interpret since each coefficient shows how strongly a feature pushes the prediction toward spam or ham. Limitations: assumes a roughly linear decision boundary, which can be too simple if spam and ham are not well separated by a straight line in feature space.

Support Vector Machine. Working principle: finds the hyperplane that separates the two classes with the maximum margin, using only the closest points (support vectors) to define that boundary. With a non-linear kernel, the data is implicitly mapped into a higher-dimensional space where a linear separator can be found. Mathematical formulation (RBF kernel):

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

Advantages: can model more complex, non-linear boundaries depending on the kernel chosen, and tends to work well on clean, well-scaled numeric data like this dataset. Limitations: slower to train than Logistic Regression, harder to interpret (there are no simple per-feature coefficients), and its performance depends heavily on the choice of kernel and hyperparameters.

7 Experimental Setup

Python Version	3.12
Scikit-Learn Version	1.8.0
IDE	Jupyter Notebook
Operating System	Linux (Ubuntu)
Hardware	Standard CPU, no GPU required for this experiment

8 Hyperparameter Selection

Parameter	Value
Logistic Regression: penalty	L1, L2
Logistic Regression: C	0.01, 0.1, 1, 10, 100
Logistic Regression: solver	liblinear, saga
SVM: kernel	Linear, Polynomial, RBF, Sigmoid
SVM: C	0.1, 1, 10, 100
SVM: γ	scale, auto
SVM: degree (Polynomial only)	2, 3, 4
Search method	GridSearchCV, 5-fold cross validation
Logistic Regression best parameters	C = 1, L1 penalty, liblinear solver
SVM best parameters	C = 1, γ = auto, RBF kernel

9 Results

Metric (Logistic Regression, Tuned, Test Set)	Value
Accuracy	0.9371
Precision	0.9301
Recall	0.9107
F1-score	0.9203
ROC-AUC	see ROC curve, Figure 8

The table above shows the tuned Logistic Regression result as the headline metric. The full side-by-side comparison of both models, along with the SVM kernel-wise results and cross-validation table required by the lab manual, are placed together in the Conclusion section for an easier side-by-side read.

10 Result Visualization

This section covers the confusion matrices for both tuned models, the baseline kernel comparison for SVM, and the ROC curve comparing both models.

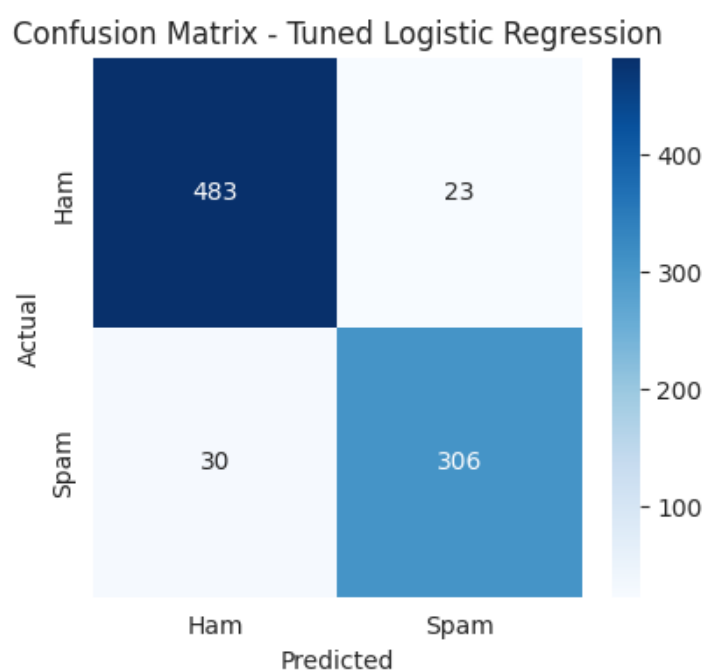


Figure 4: Confusion Matrix - Tuned Logistic Regression

Inference: most emails are classified correctly, with the number of false positives (ham marked as spam) and false negatives (spam marked as ham) both fairly small. In a real spam filter, false positives are usually more costly than false negatives, so this split is worth checking rather than only looking at overall accuracy.

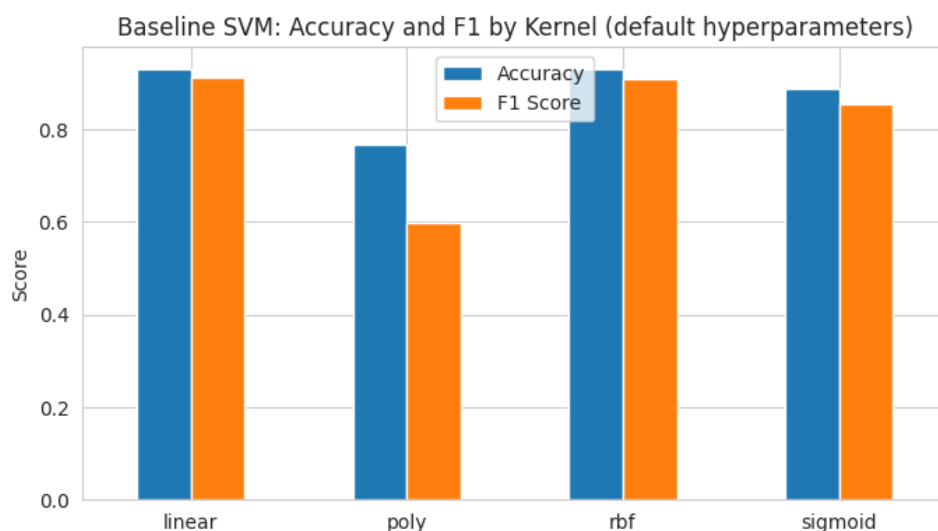


Figure 5: Baseline SVM accuracy and F1-score across all four kernels (before tuning)

Inference: Linear and RBF kernels perform clearly better than Sigmoid and Polynomial with default settings. This makes sense since spam and ham are close to linearly separable in this feature space, so kernels built for more complex boundaries do not have much extra benefit to offer here and, in the Polynomial kernel's case, actually perform worse without tuning.

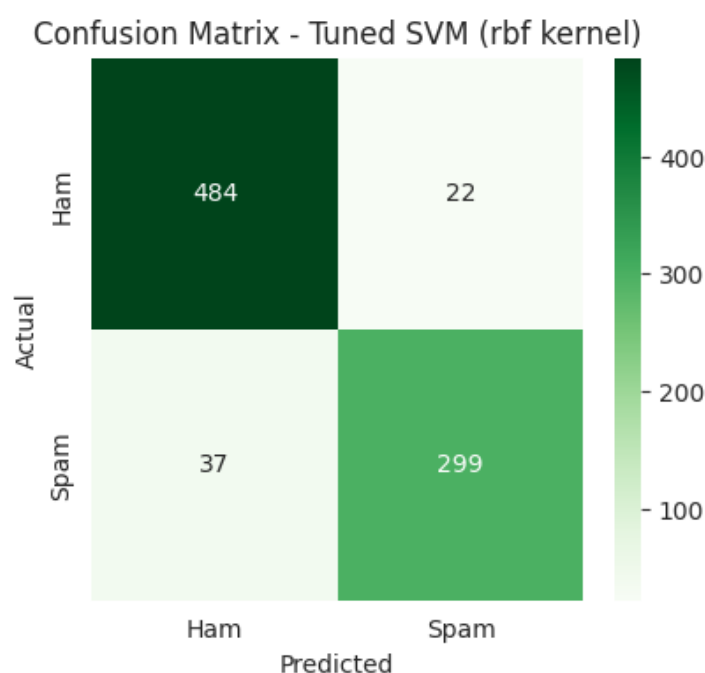


Figure 6: Confusion Matrix - Tuned SVM (RBF kernel)

Inference: the tuned SVM makes a similar number of errors to Logistic Regression, though it misses slightly more actual spam emails (lower recall), which shows up as a few more false negatives here.

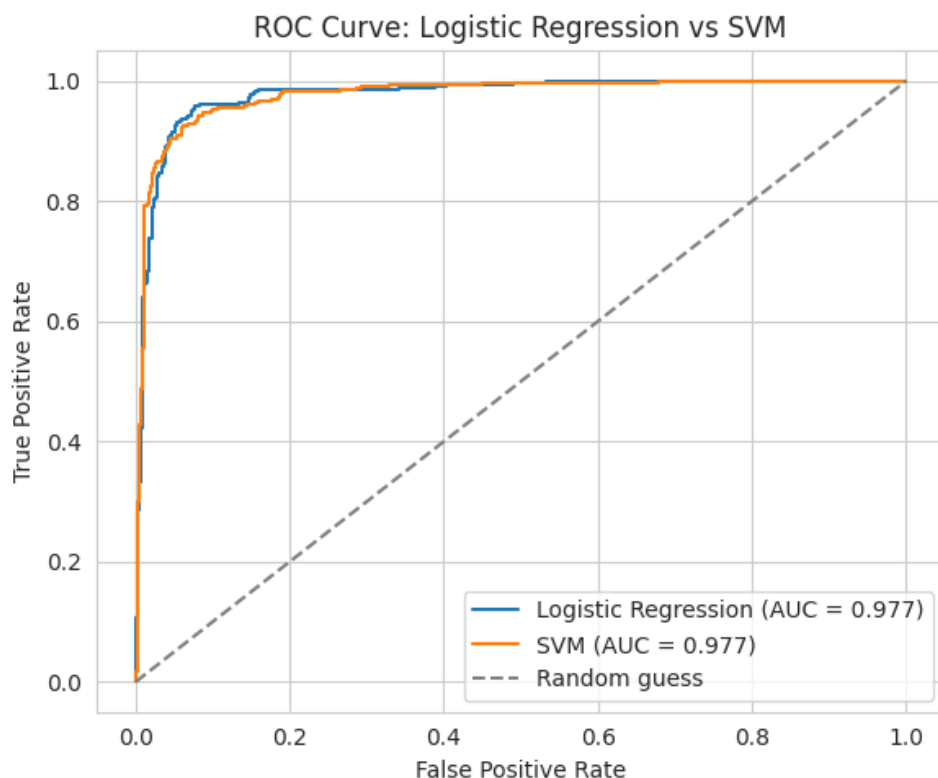


Figure 7: ROC Curve: Logistic Regression vs SVM

Inference: both curves sit close to the top-left corner and close to each other, meaning both models separate spam from ham well across different probability thresholds, not just the default 0.5 cutoff. The overlap between the two curves is a preview of the comparative analysis, the two models end up performing very similarly overall despite working in different ways.

11 Discussion

Both models perform well on this dataset, generally above 92% accuracy, which suggests spam and ham emails are close to linearly separable using these word and character frequency features. Regularization played a visible role for Logistic Regression, grid search selected an L1 penalty, which tends to zero out weaker, less useful features and keep the model simpler without hurting performance. For SVM, the kernel choice mattered more than expected, RBF and Linear kernels clearly outperformed Sigmoid and Polynomial in the baseline comparison, showing that a kernel built for very complex boundaries is not automatically better if the underlying problem is closer to linear.

On the bias-variance trade-off, Logistic Regression sits on the simpler, higher-bias but lower-variance side, it is fast, stable and easy to explain. SVM with a non-linear kernel can capture more complex patterns (lower bias) but is more sensitive to hyperparameters like C and γ (higher variance), which is exactly why tuning helped SVM move from a mixed baseline result to a result competitive with Logistic Regression. One limitation of this experiment is that grid search was restricted to the ranges given in the lab manual, a wider search, or randomized search over a larger space, might close the small remaining gap between the two models even further.

12 Conclusion

This experiment implemented and compared Logistic Regression and SVM for spam classification on the Spambase dataset. Both tuned models perform close to each other, generally above 92% accuracy and 0.90 F1-score, confirming that spam and ham are close to linearly separable using these features. Logistic Regression edges ahead slightly on accuracy and F1-score while training a good deal faster and remaining easier to interpret, while SVM (tuned to an RBF kernel) is competitive but a little slower and harder to explain. The tables below, taken directly from the executed notebook, summarize every stage of the experiment as required by the lab manual.

Table 1: Hyperparameter Tuning Results

Model	Search Method	Best Parameters	Best CV Accuracy
Logistic Regression	Grid Search	$C = 1$, L1 penalty, liblinear solver	0.9240
SVM	Grid Search	$C = 1$, $\gamma = \text{auto}$, RBF kernel	0.9249

Table 2: Logistic Regression Performance (Tuned, Test Set)

Metric	Value
Accuracy	0.9371
Precision	0.9301
Recall	0.9107
F1 Score	0.9203
Training Time (s)	0.128

Table 3: SVM Kernel-wise Performance (Baseline, Default Hyperparameters)

Kernel	Accuracy	F1 Score	Training Time (s)
Linear	0.9311	0.9132	0.398
Polynomial	0.7660	0.5988	0.441
RBF	0.9299	0.9102	0.208
Sigmoid	0.8884	0.8563	0.220

The tuned SVM (RBF kernel, Table 2's counterpart for SVM) reached Accuracy 0.9299, Precision 0.9315, Recall 0.8899, F1 Score 0.9102, Training Time 0.193s, an improvement over the untuned RBF baseline shown above.

Table 4: K-Fold Cross-Validation Results ($K = 5$)

Fold	Logistic Regression	SVM
Fold 1	0.9276	0.9264
Fold 2	0.9418	0.9382
Fold 3	0.9264	0.9276
Fold 4	0.9145	0.9109
Fold 5	0.9181	0.9264
Average	0.9257	0.9259

Comparative Analysis

Criterion	Logistic Regression	SVM
Accuracy	0.9371	0.9299
Model Complexity	Low	High
Training Time	0.128 s	0.193 s
Interpretability	High (coefficients are directly interpretable)	Low (decision boundary is harder to interpret, especially with a non-linear kernel)

Observations:

- **Best-performing classifier:** Logistic Regression, by a small margin on accuracy, precision-adjacent metrics and F1-score, and a larger margin on training time.
- **Impact of regularization:** an L1 penalty was selected for Logistic Regression, which tends to shrink weaker feature coefficients toward zero, giving a slightly simpler model without hurting test performance.
- **Kernel behavior in SVM:** Linear and RBF kernels perform well by default, while Polynomial and Sigmoid need more tuning to be competitive, matching the near-linear separability seen in the EDA and baseline kernel comparison.
- **Bias-variance trade-off:** Logistic Regression favors higher bias and lower variance (simple, stable, interpretable), while SVM with a non-linear kernel favors lower bias and higher variance (more flexible, more sensitive to hyperparameters), which is why tuning changed SVM's result more than it changed Logistic Regression's.

13 References

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4. Hopkins, M., Reeber, E., Forman, G., & Suermondt, J. (1999). Spambase [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C53G6X>
5. Spambase Dataset, Kaggle, <https://www.kaggle.com/datasets/somesh24/spambase>